

System Messages

Reminder: Click here to see the action(s) that need to be performed. [🔗 \(/config/logs/post-installation_actions.txt\)](/config/logs/post-installation_actions.txt)

Once you perform them, click here to remove this message. (/execute.php?id=AM_RM_POST) (13 occurrences, last on March 16, 09:45:46 PM)

Clear messages

📷 Allsky Settings for RPi HQ

💾 Save changes

↺ Reset to default values



☒ Restart Allsky after saving changes, if needed?

^ Daytime Settings

Capture	No	Yes	Enable to take daytime images.
Save	No	Yes	Enable to save daytime images.
Auto-Exposure	No	Yes	Enable to use daytime auto-exposure.
Max Auto-Exposure	3000		Maximum daytime Auto-Exposure time in milliseconds (1000ms = 1 sec). Click here 🔗 (/documentation/settings/allsky.html#daymaxautoexposure) to see how this setting impacts the time between exposures.
Manual Exposure	500		Manual exposure time in milliseconds (1000ms = 1 sec). Can be a fraction. If daytime Auto-Exposure is enabled, this number is used as a starting point. Click here 🔗 (/documentation/settings/allsky.html#dayexposure) to see how this setting impacts the time between exposures.
Mean Target	0.5		The target mean brightness level when Auto-Exposure is on. 1.0 is pure white. Best used when Auto-Exposure and Auto-Gain are on. 0 disables this auto-exposure mode.
Mean Threshold	0.1		When using Mean Target this specifies how close to the target the brightness should be.
Delay	30000		Delay between daytime images in milliseconds (1000ms = 1 sec).

Top

Auto-Gain	☐ No ☒ Yes	Enable to use daytime auto-gain.
Max Auto-Gain	16	Maximum gain when Auto-Gain is enabled.
Gain	1	Manually sets the light sensitivity of the camera. A high number returns a brighter image but more noise. Can normally leave at the minimum value for daytime.
Stretch Amount	0	Amount to stretch daytime images. 0 means don't stretch.
Stretch Mid Point	10	Mid point of image stretch. Lower numbers brighten darker parts of image; higher numbers brighten lighter parts. Only applies if Stretch Amount is greater than 0.
Binning	1x1 ▾	Binning level. 1x1 = OFF.
Auto White Balance	☒ No ☐ Yes	Sets auto white balance so the camera adjusts the color.
Red Balance	2.5	Red component for the white balance. When Auto White Balance is enabled, this number is a starting point.
Blue Balance	2	Blue component for the white balance. When Auto White Balance is enabled, this number is a starting point.
Frames To Skip	5	When starting Allsky during the day, skip up to this many frames while the software gets to the correct exposure.
Tuning File	 Name of the optional day camera tuning file to use. See the documentation for a description of this setting 🔗 (/documentation/settings/allsky.html#daytuningfile).	
^ Nighttime Settings		
Capture	☐ No ☒ Yes	Enable to take nighttime images.
Save	☐ No ☒ Yes	Enable to save nighttime images.
Auto-Exposure	☐ No ☒ Yes	Enable to use nighttime auto-exposure.
Max Auto-Exposure	60000	Maximum nighttime Auto-Exposure time in milliseconds (1000ms = 1 sec).
Manual Exposure	10000	Manual exposure time in milliseconds (1000ms = 1 sec). Can be a fraction. If nighttime Auto-Exposure is enabled, this number is used as a starting point.

Mean Target	0.2	The target mean brightness level when Auto-Exposure is on. 1.0 is pure white. Best used when Auto-Exposure and Auto-Gain are on. 0 disables this auto-exposure mode.
Mean Threshold	0.1	When using Mean Target this specifies how close to the target the brightness should be.
Delay	30000	Delay between nighttime images in milliseconds (1000ms = 1 sec).
Auto-Gain	☐ No ☒ Yes	Enable to use nighttime auto-gain.
Max Auto-Gain	1600	Maximum gain when Auto-Gain is enabled.
Gain	16	Manually sets the light sensitivity of the camera. A high number returns a brighter image but more noise.
Stretch Amount	0	Amount to stretch nighttime images. 0 means don't stretch.
Stretch mid point	10	Mid point of image stretch. Lower numbers brighten darker parts of image; higher numbers brighten lighter parts. Only applies if Stretch Amount is greater than 0.
Binning	1x1 ▾	Binning level. 1x1 = OFF.
Auto White Balance	☒ No ☐ Yes	Enable to use auto white balance so the camera adjusts the color. NOTE: enabling this can significantly increase the time needed to capture a nighttime image, so it's usually best to disable this at night.
Red Balance	2.5	Red component for the white balance. When Auto White Balance is enabled, this number is a starting point.
Blue Balance	2	Blue component for the white balance. When Auto White Balance is enabled, this number is a starting point.
Frames To Skip	1	When starting Allsky at night, skip up to this many frames while the software gets to the correct exposure.
Tuning File		
Name of the night camera tuning file to use.		
⤴ Both Day and Night Settings		
Days To Keep	7	Number of days of images and videos to keep on the Pi. 0 keeps all days.
Configuration File	[none]	
Configuration file to use for settings.		
Top		

Extra Arguments			Extra arguments to pass to the capture program that Allsky doesn't support.
Saturation	1		Changes the saturation level of the image. The lowest level produces a black and white image.
Contrast	1		Changes the contrast of an image.
Sharpness	1		Changes the sharpness of an image. Too sharp and images look unnatural.
Image Type	auto ▾		'auto' uses color if possible, otherwise the best mono mode supported. RAW16 only works with PNG extension.
Quality / Compression	100		JPG quality: 0-100. Higher numbers produce larger, clearer files. PNG compression: 0-9. Higher numbers produce smaller files but take longer to save.
Filename	image.jpg		Extensions allowed: .jpg or .png. WARNING: saving a .png file can take 10 or more seconds so be sure the time between successive images is long enough. To measure how long it takes your Pi to save a .png file, set the Debug Level to 4 and look in the log file.
Rotation	None ▾		Set image rotation to enable proper orientation.
Flip	None ▾		Flips the image along an axis.
Record Focus	No	Yes	Enable to have the image focus determined and output via the 'FOCUS' variable.
Consistent Delays Between Images	No	Yes	Enable to have the delays between images be a consistent length. The time between the start of exposures will always be Max Auto-Exposure + Delay .
Time Format	%Y%m%d %H:%N		Determines the format of the displayed date and time. Run <code>man 3 strftime</code> to see the options.
Temperature Units	Celsius ▾		Unit(s) to display temperature in.
Latitude	50.8763N		Camera latitude: a signed number from -90 to +90 , or an unsigned number with direction (N or S), for example: 60.7N .
Longitude	7.6172E		Camera longitude: a signed number from -180 to +180 , or an unsigned number with a direction (E or W), for example: 135.05W .
Angle	-6		Altitude of the Sun in degrees relative to the horizon at which to switch between day and night.

Top

Take Dark Frames	☐ No ☐ Yes	Enable to capture dark frames, which are used to decrease noise. See the Dark frames 🔗 (/documentation/explanations/darkFrames.html#howtousedarks) documentation on how to take dark frames.
Use Dark Frames	☐ No ☐ Yes	Enables dark frame subtraction if you have dark frames.
Locale	C.UTF-8	Your locale, used to determine what the thousands and decimal separators are. Click here 🔗 (/execute.php?ID=AM_ALLSKY_CONFIG_show_installed_locales) to see which locales are configured on your Pi.
Debug Level	1 ▾	Debug level. Use 3 while debugging. See the documentation 🔗 (/documentation/settings/allsky.html#debuglevel) for meaning of the levels.
^ Image Overlay Settings		
Overlay Method	module ▾	Determines how image overlays are done. module supports a visual editor. When set to module , the overlay settings below do NOT apply.
Show Time	☐ No ☒ Yes	Enable to display the time an image was taken on the overlay.
Show Temperature	☐ No ☒ Yes	Enable to display the camera sensor temperature on the overlay.
Show Exposure	☐ No ☒ Yes	Enable to display the exposure length on the overlay. If Auto-Exposure is set, '(auto)' will appear after the exposure time.
Show Gain	☐ No ☐ Yes	Enable to display the camera gain on the overlay. If Auto-Gain is set, '(auto)' will appear after the gain value.
Show Mean Brightness	☐ No ☐ Yes	Enable to display the calculated mean image brightness on the overlay.
Show Focus Metric	☐ No ☐ Yes	Enable to display a focus metric to help you focus the camera. This is only usefull while focusing.
Text Overlay		
	Text that appears below the optional time with the same color and size.	
Extra Text File		
	Extra Text File (enter full path to the file). Leave blank if NOT using an extra text file.	
Max Age Of Extra	0	The maximum age of the Extra Text File in seconds. After this time the contents of the file will no longer be displayed. Set to 0 to always display.
Line Height	30	The line height of the text in pixels. If increasing the font size causes the text to overlap then increase this number. Top

Text X	15	Text will begin this many pixels from the left.
Text Y	35	Text will begin this many pixels from the top.
Font Name	Simplex v	Font name for the text overlay. Used for both large and small text.
Font Color	255 0 0	Blue, Green, and Red (BGR) values of text color. NOTE: When using RAW 16 only the first two numbers are used i.e., 255 128 0 .
Small Font Color	0 0 255	BGR value of text color. NOTE: When using RAW 16 only the first two numbers are used i.e., 255 128 0 .
Font Smoothness	Antialiased v	Choose between smooth font lines (antialiased), 8-pixel, or 4 pixel connectivity (harder edges).
Font Size	7	Text Font Size. The time, if displayed, will use this size; other text displayed will be about 20% smaller.
Font Weight	1	How thick the text line should be.
Use Outline Font	No Yes	Adds a black outline to the text overlay to improve contrast.
^ Post Capture Settings		
Remove Bad Images Threshold Low	0.1	Delete images whose mean is below this value. 0 disables this check.
Remove Bad Images Threshold High	0.9	Delete images whose mean is above this value. 0 disables this check.
Create Image Thumbnails	No <input checked="" type="text" value="Yes"/>	Enable to create thumbnails of images. If you never look at them, disable this.
Thumbnail Width	100	Width of image and video thumbnails.
Thumbnail Height	75	Height of image and video thumbnails.
Image Resize Width	0	Width of resized image before cropping. 0 means don't resize. Must be an even number.
Image Resize Height	0	Height of resized image before cropping. 0 means don't resize. Must be an even number.
Image Crop Top	0	Number of pixels to crop off the top of image.
Image Crop Right	0	Number of pixels to crop off the right side of image.

Top

Image Crop Bottom

Number of pixels to crop off the bottom of image.

Image Crop Left

Number of pixels to crop off the left side of image.

^ Timelapse Settings

Daily Timelapse

Generate

☐ No☒ Yes

Enable to generate a timelapse video at the end of night.

Upload

☐ No☒ Yes

Enable to upload the timelapse video to an Allsky Website and/or remote server.

Upload
Thumbnail☐ No☒ Yes

Enable to upload the timelapse video thumbnail.

Width

Width of resized timelapse video. disables resize. Must be an even number.

Height

Height of resized timelapse video. disables resize. Must be an even number.

Bitrate

Bitrate (in K) of timelapse video. Higher values produce higher quality but larger files. Do NOT add k .

FPS

Frames Per Second (FPS) of timelapse video. Higher values will produce smoother, but faster videos.

Keep Sequence

☒ No☐ Yes

Use for debugging purposes only. Enable to keep the list of files used in creating the timelapse video.

Extra
ParametersOptional additional timelapse creation parameters. Run `ffmpeg -?` to see the options.If quality is poor or the video does not play on Apple devices, try adding `-level 3.1` .

Mini-Timelapse

Number Of
ImagesNumber of images in a mini-timelapse. disables mini-timelapse creation.

Force Creation

☒ No☐ YesCreate a mini-timelapse even if **Number Of Images** isn't reached?

Frequency

Make a mini-timelapse every X images.

If you have a slow Pi or short delays between images, set this to a higher number (i.e., not as often).

Upload

☒ No☐ Yes

Enable to upload the mini-timelapse video to an Allsky Website and/or remote server.

[Top](#)

Upload Thumbnail	☐ No ☒ Yes	Enable to upload the mini-timelapse video thumbnail.
Width	0	Width of resized mini-timelapse video. 0 disables resize. Must be an even number.
Height	0	Height of resized mini-timelapse video. 0 disables resize. Must be an even number.
Bitrate	2000	Bitrate (in K) of timelapse video. Higher values produce higher quality but larger files. Do NOT add k .
FPS	5	Frames Per Second (FPS) of mini-timelapse video.
Both Timelapses		
VCODEC	libx264	Video encoder for timelapse. Rarely changed.
Pixel format	yuv420p	Pixel format for timelapse. Rarely changed.
Log Level	Warnings + Errc	Amount of information to display while creating a timelapse. Rarely changed.
^ Keogram and Startrails Settings		
Keograms		
Generate	☐ No ☒ Yes	Enable to generate a keogram at the end of night.
Upload	☐ No ☒ Yes	Enable to upload the keogram to an Allsky Website and/or remote server.
Expand	☐ No ☒ Yes	Enable to expand keograms to the image width.
Font Name	Simplex	Font name.
Font Color	#ffffff	Font color. See the documentation 🔗 (/documentation/settings/allsky.html#keogramfontcolor) for a description of this field.
Font Size	2	Font size.
Font Line Thickness	3	Font line thickness.
Extra Parameters		
	Optional additional keogram creation parameters. Run <code>~/allsky/bin/keogram --help</code> for a list of options.	
Startrails		
Top		

Generate	☐ No ☒ Yes	Enable to generate a startrails at the end of night.
Brightness Threshold	0.49	Images with a brightness higher than this threshold will not be included in the startrails.
Upload	☐ No ☒ Yes	Enable to upload the startrails to an Allsky Website and/or remote server.
Extra Parameters		
	Optional additional startrails creation parameters. Run <code>~/allsky/bin/startrails --help</code> for a list of options.	
^ Websites and Remote Server Settings		
Resize Uploaded Images Width	0	Resized width of uploaded images. 0 means don't resize. Must be an even number.
Resize Uploaded Images Height	0	Resized height of uploaded images. 0 means don't resize. Must be an even number.
Upload Every X Images	1	How often should images be uploaded to an Allsky Website and/or remote server? 0 never uploads images, 1 uploads every frame, 2 every other frame, etc.
Display Settings	☒ No ☐ Yes	Enable to add a link on the Allsky Website to display the settings on this page.
Local Website		
Use Local Website	☒ No ☐ Yes	Enable to use the local Allsky Website. You should configure it in the Editor page first.
Days To Keep on Pi Website	0	Number of days of images and videos to keep on the Pi's Website. 0 keeps all days.
Remote Website		
Use Remote Website	☒ No ☐ Yes	Enable to use a remote Allsky Website. See the Allsky Documentation for how to install the Website.
Server Name		
	Name of server hosting the remote Allsky Website. NOT a URL.	
Website URL		
	The URL of your Allsky Website, for example: https://mywebsite.com/allsky/ . Must begin with http or https .	
Image URL		

Top

The URL of the image on your Allsky Website, for example:

`https://mywebsite.com/allsky/image.jpg`.

Must begin with `http` or `https`.

Image Directory	allsky	Name of the top-level remote Website directory where images go.
Protocol	ftp ▾	Protocol for remote Website
Port		Optional port required by server. Rarely required.
User Name		Username of the login on the remote Website server.
Password		Password of the login on the remote Website server.
FTP Commands		
	Special commands needed when connecting to an FTP server.	
SSH Public Key File		
	Path on the Pi to the SSH public key file.	
AWS CLI Directory		
	AWS CLI directory where the AWS CLI tools are installed. Often stored in <code>\${HOME}/.local/bin</code>	
S3 Bucket	allskybucket	Name of S3 bucket where files will be upload.
S3 ACL	private	S3 Access Control List (ACL).
GCS Bucket	allskybucket	Name of GCS bucket where files will be upload.
GCS ACL	private	GCS Access Control List (ACL).
Upload With Original Name	<input checked="" type="radio" value="No"/> No <input type="radio" value="Yes"/> Yes	Enable to upload images using YYYYMMDDHHMMSS naming. Rarely used.
Remote Server		
Use Remote Server	<input checked="" type="radio" value="No"/> No <input type="radio" value="Yes"/> Yes	This is a remote server NOT running the Allsky Website. See the documentation for the necessary directory layout.
Server Name		
	Name of remote server NOT hosting an Allsky Website. NOT a URL.	

Top

Image Directory		Name of the top-level remote server directory where images go.
Protocol	ftps ▾	Protocol for remote server
Port		Optional port required by remote server. Rarely required.
User Name		Username of the login on the remote server.
Password		Password of the remote server login.
FTP Commands	 Special commands needed when connecting to an FTP server.	
SSH Public Key File	 Path on the Pi to the SSH public key file.	
AWS CLI Directory	 AWS CLI directory where the AWS CLI tools are installed. Often stored in \${HOME}/.local/bin	
S3 Bucket	allskybucket	Name of S3 bucket where files will be upload.
S3 ACL	private	S3 Access Control List (ACL).
GCS Bucket	allskybucket	Name of GCS bucket where files will be upload.
GCS ACL	private	GCS Access Control List (ACL).
Upload With Original Name	☒ No ☐ Yes	Enable to upload images using YYYYMMDDHHMMSS naming.
Remote Video File Name	allsky.mp4	Optional name of the remote video file.
Remote Keogram File Name	keogram.jpg	Optional name of the remote keogram file.
Remote Startrails File Name	startrails.jpg	Optional name of the remote startrails file.
^ Allsky Map Settings		
Show on Map	☐ No ☒ Yes	Enable to have your camera appear on the Allsky Map  (https://www.thomasjacquin.com/allsky-map/). Top

Location

The location of your camera, for example: Whitehorse, YT.
This setting and the remaining ones also appear on the Allsky Website, if installed.

Owner

The owner of the camera. It can be your name, an association, or observatory, etc.

Lens

The lens you're using on your camera, for example: .

Equipment Info

Optional information on your equipment.

^ WebUI Configuration

Images Sort Order ▾

The order the images on the Images page are sorted in.
Ascending is oldest to newest. **Descending** is newest to oldest.

Show Updated Message

Disable to hide the Daytime images updated every... message in the Live View page.

Require WebUI Login

Determines if you need to login to the WebUI or not.
If your Pi is accessible on the Internet, do NOT turn this off!!.

System Page Additions

See the documentation for a description of this setting [🔗](#) (/documentation/settings/allsky.html).

^ Camera

Camera Type ▾

The type of camera you are using.
To change to a camera of a different type, select the new type then press the 'Save changes' button, THEN select the new **Camera Model**.

Camera Model ▾

The model of camera you are using.

[Top](#)